STANDARD OPERATING PROCEDURE

Single-Axis Silicone Mould Roller

Operation, cleaning and maintenance of the upgraded single-axis silicone mould rotating device

Document number	SOP-BFL-RLR-01
Revision	1.0 (first issue)
Effective date	18 October 2026
Next review	October 2027, or earlier if the device is modified
Equipment	Single-axis silicone mould roller (engraved UTM/BIOFLUID LAB, SINGLE AXIS SILICON ROLLER)
Location	Biofluid Laboratory (Dr. Zahran Lab), Department of Biomedical Engineering and Health Sciences, Faculty of Electrical Engineering, UTM Johor Bahru
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Approved by	Dr. Ahmad Zahran Md Khudzari, Senior Lecturer and laboratory head
Related documents	Visual SOP poster (Appendix A); Technical Project Report TR-BFL-RLR-01; silicone manufacturer's data sheet and SDS

	Name	Signature	Date
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Reviewed			
Approved	Dr. Ahmad Zahran Md Khudzari		

> **NOTE** Read this procedure fully before using the roller for the first time. Only users who have been trained and signed off in Section 14 may operate the device without supervision.

1 Purpose

This procedure describes how to prepare, operate, stop, clean and maintain the single-axis silicone mould roller in the Biofluid Laboratory. Its aim is to produce consistent silicone castings while protecting the user, the mould and the device. It covers the upgraded unit, which replaced the laboratory's earlier rotating device with a rail-mounted design, a labelled operator panel with ON/OFF and speed control, an RPM and status display, and a redesigned motor-to-roller hub that is clamped to the motor shaft.

2 Scope

- Applies to every user of the single-axis silicone mould roller in the Biofluid Laboratory, including staff, postgraduate students, undergraduate project students and industrial trainees.
- Covers pre-use inspection, silicone preparation, mould mounting, start-up, running, shutdown, demoulding, cleaning, routine maintenance, fault finding and emergency actions.
- Does not cover the design of new moulds, the selection of silicone grades or modifications to the device. Any modification must be approved by the laboratory head and recorded in the revision history.

3 Responsibilities

Table 1 Responsibilities

Role	Responsibilities
Laboratory head (Dr. Ahmad Zahran Md Khudzari)	Approves this SOP and its revisions. Authorises trained users. Decides when a faulty device is returned to service.
Trained operator	Follows this SOP for every run. Completes the pre-use checklist and run log. Stops the device and reports any fault, damage or near miss at once.
Laboratory technician or maintainer	Carries out the maintenance in Section 11, keeps spare hub discs and fasteners, and records repairs in the maintenance log.
New or untrained user	Uses the device only under direct supervision of a trained operator until signed off.

4 References

- Product design specification (PDS) for the Single-Axis Silicone Mould Roller Upgrade, issued by Dr. Ahmad Zahran Md Khudzari.
- Technical Project Report TR-BFL-RLR-01: Single-Axis Silicone Mould Roller Upgrade.
- Manufacturer's technical data sheet (TDS) and safety data sheet (SDS) of the silicone in use.
- UTM laboratory safety rules and the Faculty of Electrical Engineering laboratory regulations.

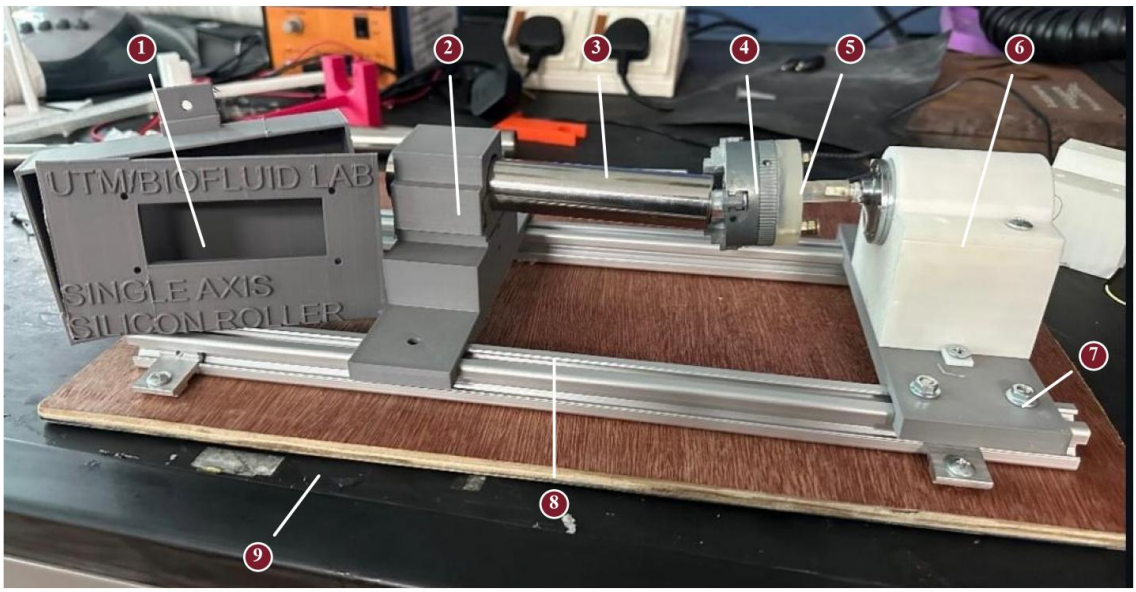
5 Definitions and Abbreviations

Table 2 Definitions and abbreviations

Term	Meaning
Roller	The polished stainless-steel tube that carries and turns the mould about one horizontal axis.
Chuck	The knurled clamp at the driven end of the roller tube.
Hub disc	The 3D-printed disc that couples the motor's 6 mm D-shaft to the chuck. Clamped to the shaft by an M3 grub screw.
Gel time	The time after mixing at which the silicone stops flowing. The mould must keep turning until this point.
Cure time	The time after which the silicone has reached handling strength and can be demoulded.
Pot life	The working time of mixed silicone before it becomes too thick to pour.
RPM	Revolutions per minute.
PDS	Product design specification.
PPE	Personal protective equipment.
SDS / TDS	Safety data sheet / technical data sheet.

6 Equipment Description

The roller is a bench device on a plywood base. An aluminium extrusion rail runs along the base and every other component clamps to it with T-nuts, so the parts can be slid along the rail and re-aligned for different moulds. From left to right the device carries the control enclosure, a 3D-printed bearing support, the stainless-steel roller tube with its knurled chuck, the hub disc, and the motor in its housing on a separate bracket (Figure 1).



1 Control enclosure with display window 2 3D-printed bearing support 3 Stainless-steel roller tube

4 Knurled chuck 5 D-bore hub disc 6 Motor housing

7 Motor bracket with T-nut clamps 8 Aluminium extrusion rail 9 Plywood base

Figure 1 Main parts of the single-axis silicone mould roller.

Table 3 Parts of the roller

No.	Part	Function and what to check
1	Control enclosure with display window	Houses the controller, the ON/OFF switch, the speed control and the display. Must be dry and closed.
2	3D-printed bearing support	Carries the free end of the roller tube. Check that it is clamped and has no cracks.
3	Stainless-steel roller tube	Carries and turns the mould. Must be clean and free of cured silicone.
4	Knurled chuck	Connects the roller tube to the hub disc. Check that its screws are tight.
5	D-bore hub disc	Couples the 6 mm D-shaft to the chuck. The M3 grub screw must bear on the shaft flat.
6	Motor housing	Encloses the drive motor. Keep silicone and liquids away from it.
7	Motor bracket with T-nut clamps	Holds the motor in line with the roller axis. Check the clamp screws.
8	Aluminium extrusion rail	Common mounting rail for all parts. Keep the slots clean.
9	Plywood base	Carries the device on the bench. Must sit flat with no rocking.

6.1 Operator controls

Table 4 Operator controls and indicators

Control or indicator	Function
ON/OFF switch (operator panel)	Switches the device on and off. OFF is the safe state for mounting, adjusting and cleaning.
Speed control knob	Sets the roller speed. Turn it fully down before switching on and before switching off.
LED display	Shows the roller speed in RPM and the operating status.
Wall-socket adapter and plug	Supplies power from a standard 230 V wall socket. Unplugging it isolates the device.

6.2 Main specifications

Table 5 Main specifications

Item	Specification
Motion	Single horizontal axis of rotation
Speed	Adjustable from the operator panel; design target up to 100 RPM (PDS item b)
Power supply	Wall-socket adapter. Check the rating label on the adapter before use
Operator panel	Dedicated ON/OFF control and speed control (PDS items c and e)
Display	LED display of RPM and operating status (PDS item f)
Drive coupling	Motor with 6 mm D-type shaft; 3D-printed hub disc Ø47 × 10 mm with an 18 mm hub, 6.2 mm D-bore and M3 × 6 grub screw
Construction	Aluminium extrusion rail on plywood base; 3D-printed PLA supports, motor seat and enclosure (PDS item g)

7 Health and Safety

WARNING Rotating parts can catch gloves, sleeves, hair and cables. Never reach near the turning chuck, hub disc or mould. Switch OFF and wait for the roller to stop before touching any part.

Table 6 Hazards and controls

Hazard	Possible harm	Control measures
Rotating roller, chuck and hub	Entanglement of fingers, gloves, hair or sleeves	Keep hands clear while running. Tie back long hair, roll up loose sleeves, remove lanyards. Switch OFF before any adjustment.
Pinch points between chuck, hub and motor housing	Pinched or cut fingers	Turn the roller by hand only with the device switched OFF and unplugged.
Mould coming loose	Mould falls, liquid silicone spills	Secure the mould and turn it one full turn by hand before switching on. Start at low speed.
Liquid silicone and catalyst	Skin and eye irritation; slippery bench	Wear nitrile gloves and safety glasses when mixing and pouring. Follow the SDS. Clean spills at once.
Electrical supply	Electric shock, fire if liquid enters the electronics	Use an undamaged adapter and cable. Keep liquids away from the enclosure. Unplug before cleaning.
Long unattended runs	Fault not noticed	Stay with the device for the first minutes of every run and check it regularly during long runs.

7.1 Personal protective equipment

- **Nitrile gloves** when handling, mixing and pouring silicone and when cleaning. Keep gloved hands away from the turning roller.
- **Safety glasses** when mixing, pouring and demoulding.
- **Laboratory coat** with sleeves that can be fastened or rolled up.
- **Closed shoes**, and long hair tied back.

8 Materials and Tools

Table 7 Materials and tools

Item	Use
Silicone (two-part) and its TDS and SDS	Casting material; the data sheet gives the mix ratio, pot life, gel time and cure time
Digital scale (0.1 g resolution)	Weighing the two parts and the mould
Mixing cups and stirring sticks	Mixing the silicone
Vacuum chamber (if available)	De-gassing the mixed silicone to remove bubbles

Item	Use
Prepared, clean mould with release agent if required	Shapes the casting
Hex keys, small screwdriver	Checking clamps, the hub grub screw and chuck screws
Handheld tachometer (optional)	Checking the displayed RPM during commissioning or after repair
Dry lint-free wipes and isopropyl alcohol	Cleaning uncured silicone from the device
Run log sheet (Section 13)	Recording every run

9 Procedure

Carry out the steps in order. Tick each check on the pre-use checklist (Form A in Section 13). If any check fails, do not use the device: tag it as out of service and report it to the laboratory head.

9.1 Pre-use inspection (device OFF and unplugged)

Table 8 Pre-use inspection

Step	Action	Acceptance check
1.1	Clear the bench around the device. Place it on a flat, dry surface with the cable routed away from the roller.	Base flat, no rocking; nothing within reach of the roller
1.2	Inspect the adapter, cable and plug.	No cuts, exposed wires or heat damage
1.3	Check the rail clamps of the bearing support, motor bracket and enclosure with a hex key.	No movement of any part along the rail
1.4	Check the hub disc: the M3 grub screw bears on the shaft flat and the two screws to the chuck are tight.	No play between hub, shaft and chuck
1.5	Turn the roller slowly by hand through two full turns.	Turns freely, runs true, no rubbing, no noise
1.6	Check the roller tube and chuck for cured silicone or dirt.	Clean surfaces
1.7	Set the ON/OFF switch to OFF and turn the speed knob fully down.	Switch OFF, speed at minimum

9.2 Preparing the silicone

Table 9 Silicone preparation

Step	Action	Acceptance check
2.1	Read the silicone TDS. Note the mix ratio, pot life, gel time and cure time on the run log.	Values recorded
2.2	Calculate the silicone mass needed for the required wall thickness, and add a small allowance for losses.	Mass written on the run log
2.3	Wearing gloves and glasses, weigh both parts into a clean cup on the scale.	Ratio within the TDS tolerance
2.4	Mix thoroughly, scraping the sides and bottom of the cup.	Uniform colour, no streaks
2.5	De-gas in the vacuum chamber if available, within the pot life.	Bubbles removed

9.3 Filling and mounting the mould

Table 10 Filling and mounting the mould

Step	Action	Acceptance check
3.1	Pour the measured silicone into the mould and close it. Seal the joints and the pour opening.	No leaks when the mould is tilted
3.2	Wipe any silicone off the outside of the mould.	Outside of mould clean and dry
3.3	Fit the mould on the roller and secure it so that it turns with the roller without slipping.	Mould centred on the roller axis and secure
3.4	Turn the roller one full turn by hand.	Mould clears the bearing support, rail, enclosure and motor housing

CAUTION Start each run within the pot life of the silicone. If the silicone has started to thicken before the roller is running, the coating will be uneven.

9.4 Start-up and speed setting

Table 11 Start-up

Step	Action	Acceptance check
4.1	Check that the ON/OFF switch is OFF and the speed knob is fully down, then plug the adapter into the wall socket.	Nothing moves when plugged in
4.2	Switch ON at the operator panel.	Display lights up and shows the status
4.3	Turn the speed knob up slowly and watch the RPM on the display until the set speed for this mould is reached.	Smooth rotation, displayed RPM steady
4.4	Watch the first few turns closely.	No slipping, wobble, knocking or leaking

Choosing the speed. The right speed depends on the mould and the silicone and should be written on the mould's casting sheet once it has been found. A useful reference is the speed at which the centrifugal acceleration at the mould's inner wall equals gravity:

$$n_c = (60 / 2\pi)\sqrt{(g / r)} \approx 29.9 / \sqrt{r} \quad (1)$$

where n_c is the reference speed in RPM, g = 9.81 m/s^2 and r is the inner radius of the mould cavity in metres. Below n_c the liquid silicone tumbles and flows over the wall; well above n_c it is held against the wall as an even film around the circumference but can no longer flow along the mould. Start at a low speed and raise it only if the coating is uneven.

Table 12 Reference speed for different mould radii

Mould inner radius r (mm)	10	20	30	50	75	100
Reference speed n_c (RPM)	299	211	173	134	109	95

9.5 During the run

- Stay with the device for at least the first five minutes of every run, and check it at regular intervals afterwards.
- Keep the RPM steady. Do not change the speed sharply, because this can cause the silicone to slosh and leave ridges.
- If the mould slips, wobbles, knocks or leaks, turn the speed down, switch OFF and correct the fault before restarting.
- Keep the device turning until the gel time on the TDS has passed, so the silicone cannot run back into a pool.
- Do not cover the control enclosure, and keep liquids and tools away from it.

9.6 Shutdown and demoulding

Table 13 Shutdown and demoulding

Step	Action	Acceptance check
6.1	Turn the speed knob down to minimum.	Roller slows and stops
6.2	Switch OFF at the operator panel, then unplug the adapter from the wall socket.	Display off, device isolated
6.3	Remove the mould from the roller.	Mould handled level
6.4	Leave the mould until the cure time on the TDS has passed, then open it and remove the casting.	Casting tack-free and fully formed
6.5	Inspect the casting for thin spots, bubbles or thick pools, and record the result in the run log.	Result recorded

9.7 Cleaning

- Clean with the device switched OFF and unplugged.
- Wipe uncured silicone off the roller tube, chuck, hub and rail with dry wipes, then with a little isopropyl alcohol. Peel off cured silicone; do not scrape the roller with metal tools.
- Do not pour or spray liquid onto the device, and keep cleaning fluids away from the control enclosure and motor housing.
- Leave the speed knob at minimum and the switch OFF, and store the adapter cable tidily.

10 Emergency Procedures

Table 14 Emergency actions

Situation	Immediate action
Entanglement or any risk of injury	Switch OFF at the panel, then unplug at the wall socket. Do not try to stop the roller by hand. Give first aid and report the incident.
Mould comes loose or silicone spills	Turn the speed down and switch OFF. Unplug. Contain the spill with wipes; keep it away from the enclosure. Clean as in 9.7.
Smoke, burning smell or sparking	Unplug at the wall socket at once. Do not use water on electrical equipment. Use a CO_2 extinguisher if there is a fire and raise the alarm.
Silicone in the eyes	Rinse with clean water for at least 15 minutes and seek medical advice. Follow the SDS.
Silicone on the skin	Wipe off and wash with soap and water. Follow the SDS.

WARNING After any emergency, tag the device "OUT OF SERVICE" and report to the laboratory head. Do not use it again until it has been inspected and released.

11 Routine Maintenance

Table 15 Maintenance schedule

Interval	Task	By
Before every run	Pre-use inspection in 9.1	Operator
After every run	Clean the roller, chuck and rail as in 9.7	Operator
Monthly	Check all rail T-nut clamps, chuck screws and the hub grub screw for tightness; check the adapter, cable and plug	Technician
Monthly	Check the hub disc and the 3D-printed parts for cracks, wear or deformation; replace if damaged (Appendix B)	Technician
Every 6 months	Check the displayed RPM against a handheld tachometer at low, medium and high settings and record the readings	Technician
Yearly	Review this SOP and the run and maintenance logs	Laboratory head

12 Troubleshooting

Table 16 Troubleshooting guide

Problem	Likely cause	Action
Display does not light when switched ON	Adapter not plugged in, wall socket off, loose connection	Check the socket and plug. If still dead, unplug and report; do not open the enclosure.
Display on but roller does not turn	Speed knob at minimum; hub loose on the shaft	Raise the speed slowly. If the motor runs but the roller does not, switch OFF, unplug and tighten the grub screw.
Roller turns in steps or stalls	Mould too heavy or off-centre; bearing support misaligned	Switch OFF. Re-centre the mould and re-align the bearing support on the rail.
Knocking or wobble	Hub screws loose; mould not centred; rail clamp loose	Switch OFF, unplug, check every clamp and screw, re-centre the mould.
Displayed RPM does not match the expected speed	Display needs checking	Check with a tachometer and report the readings to the technician.

Problem	Likely cause	Action
Uneven wall thickness in the casting	Speed too low or too high; run stopped before gel time; mould not level	Adjust the speed as in 9.4, run for the full gel time and level the device.
Bubbles in the casting	Silicone not de-gassed; mixed too fast	Mix slowly and de-gas before pouring.

13 Records

Keep the forms below in the device folder next to the roller. Completed forms are kept for at least two years.

Form A: Pre-use checklist

Date	Operator	Base & cable OK	Clamps tight	Hub tight	Turns freely	Switch OFF, speed min	Sign

Form B: Run log

Date	Operator	Mould ID	Silicone & batch	Mass (g)	Speed (RPM)	Start / stop	Result / remarks

14 Training and Competency

1. Read this SOP, the silicone SDS and the visual SOP poster.
2. Watch a trained operator complete one full run, from inspection to cleaning.
3. Complete one full run under supervision, including the pre-use checklist and the run log.
4. Be signed off by the laboratory head or a person nominated by them in the table below.

Name	Matric / staff no.	Trained by	Date	Signature

15 Revision History

Rev.	Date	Description	By
1.0	18 October 2026	First issue for the upgraded single-axis silicone mould roller	M. Alamin A. Y. Ahmed

Appendix A Visual SOP

Print this page in colour on A4 and display it next to the device. It summarises Sections 7 to 10.



Figure 2 Visual SOP poster for the single-axis silicone mould roller.

Appendix B Replacing the Hub Disc

The hub disc is a 3D-printed part and can be reprinted from its STL file if it cracks or wears. Its main dimensions are given in the technical report (TR-BFL-RLR-01, Table 3).

Step	Action
B1	Switch OFF and unplug the device.
B2	Remove the two screws between the hub disc and the chuck. Loosen the M3 grub screw and slide the hub off the motor shaft.
B3	Print a new hub in PLA, back face down (ring segments on the bed). Enable supports if the Ø29 mm bridge of the centre hollow sags.
B4	Drop the M3 nut into the slot on top of the hub before fitting the shaft.
B5	Slide the hub onto the 6 mm D-shaft with the bore flat lined up with the shaft flat. Turn the M3 × 6 grub screw in from the side until it bears on the flat.
B6	Turn the roller by hand to check it runs true, then fit and tighten the two screws to the chuck.
B7	Carry out the pre-use inspection in 9.1 and record the replacement in the maintenance log.